# *Topics in Error-Correcting Codes*

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*Lecture 12: Introduction to LCL Properties of Random Linear Codes and Subspace-Design Codes*

Our starting goal is to understand list decoding near the radius $1 - R$. Throughout this module the alphabet is large: its size may depend on the rate and the desired gap to $1 - R$. Today we begin with random codes, isolate the linear constraints behind their analysis, and ask for explicit codes that obey the same constraint-counting principle. This leads to *local coordinate-wise linear* (LCL) properties and *subspace-design codes*.

## *Reminder: List Decoding*

**Definition 1** (List decoding)**.** For a code $\mathcal{C} \subseteq \Sigma^n$ and a received word $y \in \Sigma^n$, let

$$\mathcal{L}(y, \rho) = \{c \in \mathcal{C} : \Delta(c, y) \leq \rho\}.$$

The code is $(\rho, L)$*-list-decodable* if $|\mathcal{L}(y, \rho)| \leq L$ for every $y \in \Sigma^n$.

We are asking for a combinatorial bound on the number of plausible codewords; this definition does not assert that the list can be found efficiently. Fix $0 < R < 1$ and $0 < \varepsilon < 1 - R$, and put

$$\rho = 1 - R - \varepsilon.$$

Over a sufficiently large alphabet, a completely random rate-$R$ code is $(\rho, O(1/\varepsilon))$-list-decodable with high probability.

> **Motivating question.** Does a random *linear* code have the same list-decoding guarantee?

We will see why the elementary random-code proof does not immediately answer this question, and what it does give us for linear codes.

## *The Random-Code Argument*

Let $B_q(y, \rho) = \{x \in \mathbb{F}_q^n : \Delta(x, y) \leq \rho\}$. A useful large-alphabet estimate is

$$|B_q(y, \rho)| = \sum_{j=0}^{\lfloor \rho n \rfloor} \binom{n}{j} (q-1)^j \leq 2^n q^{\rho n}. \tag{1}$$

The factor $2^n$ counts possible error locations. The factor $q^{\rho n}$ bounds the number of assignments at those locations.

**Theorem 2** (Completely random codes)**.** *Let $k = Rn$ be an integer and choose $\mathcal{C}$ uniformly among the $q^k$-element subsets of $\mathbb{F}_q^n$. For any positive integer $t$,*

$$\Pr[\exists y : \ |\mathcal{L}(y,\rho)| \geq t] \leq 2^{nt} q^{n(1-t\varepsilon)}.$$

*In particular, if $q \geq 2^{2/\varepsilon}$ and $t = \lceil 4/\varepsilon \rceil$, the failure probability is at most $q^{-n}$. Thus the code is $(1-R-\varepsilon, t-1)$-list-decodable with high probability.*

*Proof.* Label the $q^k$ randomly chosen codewords. Fix a received word $y$ and $t$ distinct labels. The probability that all these codewords lie in $B_q(y,\rho)$ is at most

$$\left(\frac{|B_q(y,\rho)|}{q^n}\right)^t \leq \left(2^n q^{-(R+\varepsilon)n}\right)^t.$$

For independent uniform codewords, the product formula follows from independence. Sampling without replacement only decreases this probability: after a successful draw, the fraction of remaining words in the ball cannot increase.

There are $q^n$ choices of $y$ and at most $q^{kt}$ choices of the labels. A union bound gives

$$q^n \cdot q^{kt} \cdot \left(2^n q^{-(R+\varepsilon)n}\right)^t = 2^{nt} q^{n(1-t\varepsilon)}.$$

If $\log_q 2 \leq \varepsilon/2$, this is at most $q^{n(1-t\varepsilon/2)}$, which is at most $q^{-n}$ for the stated $t$. □

### *Where linearity changes the proof*

Generate a random linear code using a uniformly random matrix $G \in \mathbb{F}_q^{k\times n}$, with

$$\mathcal{C} = \{uG : u \in \mathbb{F}_q^k\}.$$

If $u_1, \ldots, u_t$ are linearly independent messages, then $u_1G, \ldots, u_tG$ are independent uniform words.

But distinct messages need not be linearly independent. For instance,

$$(u_1 + u_2)G = u_1G + u_2G.$$

Consequently, we cannot multiply the single-word probabilities for an arbitrary list of distinct codewords.

The same independence argument already bounds the dimension of the span of a close-by list by $O(1/\varepsilon)$. We can do better: translating by one codeword reveals the more informative scale $(1-R)/\varepsilon$.

The random generator has full row rank except with probability less than $q^{k-n}$: union-bound the events $uG = 0$ over nonzero $u$. Conditioned on full rank, it generates a uniform $k$-dimensional code. Our bounds use the unconditioned model; conditioning changes the failure bounds by at most a factor $1/(1-q^{k-n})$.

## *A Small Affine Span for Random Linear Codes*

Recall that $c_0, \ldots, c_{t-1}$ are *affinely independent* if $c_1 - c_0, \ldots, c_{t-1} - c_0$ are linearly independent. Their affine span then has dimension $t-1$.

**Theorem 3** (Bounding affine independence)**.** *For the random linear code above,*

$$\Pr[\textit{some } \mathcal{L}(y,\rho) \textit{ contains } t \textit{ affinely independent words}] \leq 2^{nt} q^{n(1-R-t\varepsilon)}.$$

*In particular, for every integer $t$ with $t\varepsilon > 1 - R$, this probability is exponentially small in $n$ over a sufficiently large field. With high probability every nonempty list has affine dimension at most $t - 2$ and linear-span dimension at most $t - 1$.*

*Proof.* Suppose $c_0, \dots, c_{t-1}$ are affinely independent and each is within distance $\rho$ of the same word $y$. Translate by $c_0$:

$$d_0 = 0, \qquad d_j = c_j - c_0 \quad (1 \le j < t), \qquad z = y - c_0.$$

The $d_j$ are still codewords, and $d_1, \dots, d_{t-1}$ are linearly independent. Crucially, we need only choose the messages for these $t - 1$ differences; we do not need to choose $c_0$.

For each $j \in \{0, \dots, t-1\}$ choose an agreement set $S_j \subseteq [n]$ of size at least $(1 - \rho)n$ on which $d_j$ equals $z$. There are at most $2^{nt}$ choices of all these sets. At coordinate $i$, put

$$J_i = \{j : i \in S_j\}.$$

All the symbols $d_{j,i}$ for $j \in J_i$ must be equal. Among the $t - 1$ variables $d_{1,i}, \dots, d_{t-1,i}$, these impose exactly

$$a_i = \max\{|J_i| - 1, 0\}$$

independent linear constraints. If $0 \in J_i$, the other symbols must all be zero. If $0 \notin J_i$, choose one of the symbols as a reference and equate the others to it. Both cases give $|J_i| - 1$ constraints when $J_i$ is nonempty.

Thus the received symbol $z_i$ has disappeared: we only need equalities among codeword symbols. The total number of constraints is

$$\begin{aligned} \sum_{i=1}^{n} a_i &\ge \sum_{i=1}^{n} (|J_i| - 1) = \sum_{j=0}^{t-1} |S_j| - n \\ &\ge t(1 - \rho)n - n = \big(t(R + \varepsilon) - 1\big)n. \end{aligned}$$

For any fixed independent messages $u_1, \dots, u_{t-1}$, their encodings have uniform, independent columns in $\mathbb{F}_q^{t-1}$. Hence they obey these constraints with probability $q^{-\sum_i a_i}$. There are at most $q^{k(t-1)}$ choices of these messages. Union-bounding over messages and agreement sets gives

$$2^{nt} q^{k(t-1)} q^{-n(t(R+\varepsilon)-1)} = 2^{nt} q^{n(1-R-t\varepsilon)}.$$

Every independent tuple of codewords has independent message preimages, so this covers every possible witness. □

*The quantitative conclusion.* Write $\eta = t\varepsilon - (1 - R) > 0$. It suffices to take $q \ge 2^{2t/\eta}$ to make the failure probability at most $q^{-\eta n/2}$. In particular, taking

$$t = \left\lfloor \frac{1 - R}{\varepsilon} \right\rfloor + 1$$

shows, for sufficiently large $q = q(R, \varepsilon)$, that every close-by list contains at most $\lfloor (1 - R)/\varepsilon \rfloor$ linearly independent codewords. Its affine dimension is at most $\lfloor (1 - R)/\varepsilon \rfloor - 1$.

This is substantial structure, but it does not yet give a list of size $O(1/\varepsilon)$. An affine space of dimension $d$ has $q^d$ points, so the conclusion so far is only a list-size bound $q^{O((1-R)/\varepsilon)}$. We still need to control the number of close-by codewords *inside* the small affine space.

## *Comparison with the Generalized Singleton Bound*

The appearance of $(1-R)/\varepsilon$ is suggestive: this is also the scale forced on the *number* of codewords by the generalized Singleton bound.

**Theorem 4** (Generalized Singleton bound)**.** *For a family of positive asymptotic rate $R$ that is $(\rho, L)$-list-decodable with fixed $L$,*

$$\rho \le \frac{L}{L+1}(1-R).$$

*Consequently, at radius $\rho = 1 - R - \varepsilon$ one must have*

$$\boxed{L \ge \frac{1-R}{\varepsilon} - 1.}$$

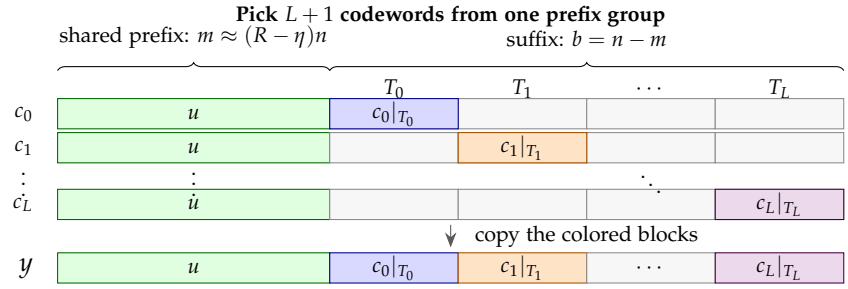
*Proof.* We will find $L+1$ distinct codewords with a long common prefix, then build a received word that also agrees with each of them on its own piece of the remaining coordinates.

**Pick the codewords by pigeonhole.** Write $Q = |\Sigma|$ and fix an arbitrarily small $0 < \eta < R$. Group the $Q^{Rn}$ codewords according to their first $m = \lfloor (R-\eta)n \rfloor$ symbols. There are only $Q^m$ possible prefixes, so some group contains at least

$$\frac{Q^{Rn}}{Q^m} \ge Q^{\eta n} > L$$

codewords for sufficiently large $n$. Pick any $L+1$ distinct codewords $c_0, \dots, c_L$ from this group. They all begin with the same string $u$. This is why we take the prefix slightly shorter than $Rn$.

**Build the received word from their suffixes.** Split the remaining $b = n - m$ coordinates into $L+1$ blocks $T_0, \dots, T_L$, with sizes differing by at most one. Let $y$ begin with $u$; on block $T_j$, copy $c_j$.




Each $c_j$ agrees with $y$ on the green prefix and its own colored block $T_j$.

**Count the possible errors.** Each $c_j$ agrees with $y$ on at least $m + \lfloor b/(L+1) \rfloor$ coordinates. Thus, simultaneously for all $j$,

$$\Delta(c_j, y) \le \frac{b - \lfloor b/(L+1) \rfloor}{n} \le \frac{L}{L+1}(1 - R + \eta) + \frac{2}{n}.$$

If $\rho > \frac{L}{L+1}(1-R)$, choose $\eta$ small enough and then $n$ large enough that the right-hand side is less than $\rho$. All $L+1$ chosen codewords then lie in the same decoding ball, contradicting $(\rho, L)$-list-decodability.

Finally, substituting $\rho = 1 - R - \varepsilon$ gives $(L+1)\varepsilon \ge 1 - R$, or $L \ge (1-R)/\varepsilon - 1$. □

## *What Did the Proof Actually Use?*

We counted independent linear constraints *at each coordinate*. The particular constraints happened to be equalities among symbols, but the calculation applies to any prescribed linear constraints.

Local profiles and the LCL framework were introduced by Levi, Mosheiff, and Shagrithaya (LMS).

**Definition 5** (Local profile: first formulation)**.** An $L$-local coordinate-wise linear (LCL) profile is a tuple of linear maps

$$\mathcal{V} = (V_1, \dots, V_n), \qquad V_i : \mathbb{F}_q^L \longrightarrow \mathbb{F}_q^L.$$

A linear code $\mathcal{C} \subseteq \mathbb{F}_q^n$ *contains* this profile if there is a matrix $M \in \mathbb{F}_q^{L \times n}$ of row rank $L$ whose rows belong to $\mathcal{C}$ and whose $i$th column satisfies

$$V_i M_{*,i} = 0 \qquad (i \in [n]).$$

Only the kernels of the maps matter. The number of constraints at coordinate $i$ is

$$r_i = \operatorname{rank} V_i = L - \dim \ker V_i.$$

**Proposition 6** (Counting a fixed profile)**.** *For a random generator $G \in \mathbb{F}_q^{k \times n}$ with $k = Rn$,*

$$\Pr[\mathcal{C} \text{ contains } \mathcal{V}] \le q^{RnL - \sum_i (L - \dim \ker V_i)} = q^{-(1-R)nL + \sum_i \dim \ker V_i}.$$

*Proof.* Any witness is $M = UG$ for some full-row-rank $U \in \mathbb{F}_q^{L \times k}$. There are at most $q^{kL}$ choices for $U$. Write $\mathrm{Enc}_i$ for the $i$th column of $G$. For fixed $U$, each $U\,\mathrm{Enc}_i$ is uniform in $\mathbb{F}_q^L$, independently over $i$. Therefore

$$\Pr[V_i U\,\mathrm{Enc}_i = 0] = \frac{|\ker V_i|}{q^L} = q^{-(L - \dim \ker V_i)}.$$

Independence across coordinates gives the product

$$\begin{aligned} \Pr[V_i U\,\mathrm{Enc}_i = 0 \text{ for all } i \in [n]] &= \prod_{i=1}^{n} \Pr[V_i U\,\mathrm{Enc}_i = 0] \\ &= \prod_{i=1}^{n} q^{-(L - \dim \ker V_i)} \\ &= q^{-\sum_{i=1}^{n} (L - \dim \ker V_i)}. \end{aligned}$$

If $\mathcal{C}$ contains the profile, these constraints hold for at least one full-row-rank $U$. Union-bounding over these matrices,

$$\begin{aligned}\Pr[\mathcal{C} \text{ contains } \mathcal{V}] &\le \sum_{\substack{U \in \mathbb{F}_q^{L\times k} \\ \operatorname{rank} U = L}} \Pr[V_i U \operatorname{Enc}_i = 0 \text{ for all } i \in [n]] \\ &\le q^{kL} \cdot q^{-\sum_{i=1}^{n}(L - \dim\ker V_i)} \\ &= q^{RnL - \sum_{i=1}^{n}(L - \dim\ker V_i)},\end{aligned}$$

where the last equality uses $k = Rn$. □

Thus a fixed profile is unlikely to occur when

$$\frac{1}{n}\sum_{i=1}^{n}(L - \dim\ker V_i) > RL.$$

For example, exceeding $RL$ by $\delta > 0$ gives probability at most $q^{-\delta n}$. A large kernel allows more symbols and therefore imposes *fewer* constraints.

> **Too many constraints are bad:** $kL = RnL$ message parameters cannot typically satisfy more than $RnL$ independent local constraints.

A property forbidding specified local profiles is the basic kind of LCL property we will study. The distinction between *a fixed* profile and *all* profiles is crucial. For a family $\mathcal{P}$ of profiles, we must also pay for the size of that family in the union bound. In the list-decoding proof, this was the factor $2^{nt}$ for agreement patterns.

*Can a code avoid every overconstrained profile?*

For scalar alphabets, this is too much to ask. Take any $L$ independent codewords and put them in the rows of $M$. At coordinate $i$, write $v_i = M_{*,i} \in \mathbb{F}_q^L$. Choose $V_i$ with

$$\ker V_i = \operatorname{span}\{v_i\}.$$

Then $V_i v_i = 0$, while $\dim\ker V_i \le 1$ and $\operatorname{rank} V_i \ge L - 1$. Hence this profile has at least $n(L-1)$ constraints and is nevertheless contained in the code. Whenever

$$L - 1 > RL \quad\Longleftrightarrow\quad L > \frac{1}{1-R},$$

it imposes more than $RnL$ constraints. This applies whenever the code has dimension at least $L$.

The profile here was chosen *after seeing the code*. There is no contradiction with the small probability of containing any one profile fixed in advance. It does rule out a universal avoidance property for scalar codes at these localities.

### *Vector Symbols Remove the Obstruction*

Now let $\mathcal{C} \subseteq (\mathbb{F}_q^s)^n$ be an $\mathbb{F}_q$-linear code. Such a code is also called *additive*; we do not require linearity over an extension field of size $q^s$. Choose independent codewords $c_1, \dots, c_L$. Their symbols at coordinate $i$ form a block

$$M_i = \begin{pmatrix} c_{1,i} \\ \vdots \\ c_{L,i} \end{pmatrix} \in \mathbb{F}_q^{L \times s}.$$

Each row displayed here is a vector in $\mathbb{F}_q^s$. Apply the same map $V_i$ separately to each of the $s$ columns:

$$V_i M_i = 0.$$

Equivalently, the column space of $M_i$ must lie in $\ker V_i$. The global $L \times sn$ matrix still has row rank $L$.

For scalar symbols, a local column spans at most one dimension. With vector symbols, the $s$ columns can span all of $\mathbb{F}_q^L$ when $s \geq L$; then $V_i M_i = 0$ forces $V_i = 0$. The automatic $L - 1$ constraints per coordinate have disappeared. We will be especially interested in $s \gg L$.

The random-code calculation has the same normalized threshold. For a random additive code with generator $G \in \mathbb{F}_q^{k \times sn}$ and rate $R = k/(sn)$, a fixed profile has probability at most

$$q^{kL - s\sum_i \operatorname{rank} V_i} = q^{s(RnL - \sum_i \operatorname{rank} V_i)}.$$

Here each local equation is imposed on all $s$ entries. Can an explicit code enforce this constraint budget *simultaneously for every profile*? Subspace-design codes express exactly this requirement, with a small relaxation of the threshold $R$.

### *Subspace-Design Codes*

For a linear subspace $\mathcal{A} \subseteq \mathcal{C}$, define

$$\mathcal{A}_i = \{a \in \mathcal{A} : a_i = 0\}.$$

Thus $\mathcal{A}_i$ is the kernel of the coordinate-evaluation map from $\mathcal{A}$ to $\mathbb{F}_q^s$. All dimensions below are over $\mathbb{F}_q$.

**Definition 7** (Subspace-design code)**.** Let $\tau : \mathbb{N} \to [0, 1]$ be nondecreasing. An additive code $\mathcal{C} \subseteq (\mathbb{F}_q^s)^n$ is a *$\tau$-subspace-design code* if, for every $r \geq 1$ and every subspace $\mathcal{A} \subseteq \mathcal{C}$ of dimension at most $r$,

$$\boxed{\frac{1}{n}\sum_{i=1}^{n} \dim \mathcal{A}_i \leq \tau(r) \dim \mathcal{A}.} \tag{2}$$

We want $\tau(r) \approx R$ for the dimensions $r$ of interest. For a one-dimensional space $\mathcal{A} = \operatorname{span}\{c\}$, the dimension of $\mathcal{A}_i$ is 1 exactly when $c_i = 0$. Thus the $r = 1$ condition

says that every nonzero codeword is zero on at most a $\tau(1)$ fraction of positions, or equivalently that the relative distance is at least $1 - \tau(1)$. The Singleton bound therefore gives $\tau(r) \geq R - 1/n$ for every $r \geq 1$.

### *Unpacking the constraint budget*

Take a basis $c_1, \ldots, c_L$ of $\mathcal{A}$, and form the blocks $M_i$ as above. The coefficient map identifies

$$\mathcal{A}_i \cong \{a \in \mathbb{F}_q^L : a^{\mathsf{T}} M_i = 0\}.$$

In particular,

$$\dim \mathcal{A}_i = L - \operatorname{rank} M_i.$$

If $V_i M_i = 0$, every row of $V_i$ belongs to this coefficient kernel, so

$$\operatorname{rank} V_i \leq \dim \mathcal{A}_i.$$

The subspace-design property now gives

$$\boxed{\sum_{i=1}^{n} \big(L - \dim(\ker(V_i))\big) \leq nL\tau(L).}$$

Thus no independent $L$-tuple of codewords can realize a profile whose constraint count exceeds $nL\tau(L)$.

Conversely, for each $i$ choose the rows of $V_i$ to span the entire coefficient kernel $\{a : a^{\mathsf{T}} M_i = 0\}$. Then $V_i M_i = 0$ and $\operatorname{rank} V_i = \dim \mathcal{A}_i$. Hence not containing any $L' \leq L$ LCL profile $(V_1, \ldots, V_n)$ such that $\sum L' - \dim \dim \ker(V_i) > RnL'$, implies the subspace-design condition. For subspaces of dimension $L \leq r$, use $\tau(L) \leq \tau(r)$.

## *Folded Reed-Solomon Codes*

Recall the $s$-folded Reed–Solomon code from Lecture 11:

$$P \longmapsto \big((P(\alpha_i), P(\alpha_i\gamma), \ldots, P(\alpha_i\gamma^{s-1}))\big)_{i=1}^{n}, \qquad \deg P < k.$$

Assume the $sn$ evaluation points are distinct and nonzero, and that $\gamma \in \mathbb{F}_q^*$ has multiplicative order greater than $k$. For example, take a primitive $\gamma$ with $q - 1 \geq sn$ and $q - 1 > k$, and $\alpha_i = \gamma^{s(i-1)}$. The rate over the alphabet $\mathbb{F}_q^s$ is $R = k/(sn)$.

**Theorem 8** (FRS codes are subspace-design codes; proof next class)**.** *For every $1 \leq r \leq s$ and every linear subspace $\mathcal{A} \subseteq \mathcal{C}$ of dimension at most $r$,*

$$\frac{1}{n} \sum_{i=1}^{n} \dim \mathcal{A}_i \leq \frac{sR}{s - r + 1} \dim \mathcal{A}.$$

*Thus we may take*

$$\tau(r) = \min\left\{1, \frac{sR}{s - r + 1}\right\} \quad (1 \leq r \leq s), \qquad \tau(r) = 1 \quad (r > s).$$

For $1 \leq r \leq s$,

$$\frac{sR}{s - r + 1} = R\left(1 + \frac{r - 1}{s - r + 1}\right).$$

Taking $s \geq r - 1 + R(r-1)/\eta$ gives $\tau(r) \leq R + \eta$. At $r = 1$, the theorem recovers the root-counting bound for FRS distance.

*Next class.* We will formalize the local-profile and subspace-design framework, and prove that folded Reed-Solomon codes satisfy the stated subspace-design bound.